Delayed, Cancelled, On Time, Boarding, ... Flying in the USA

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Data

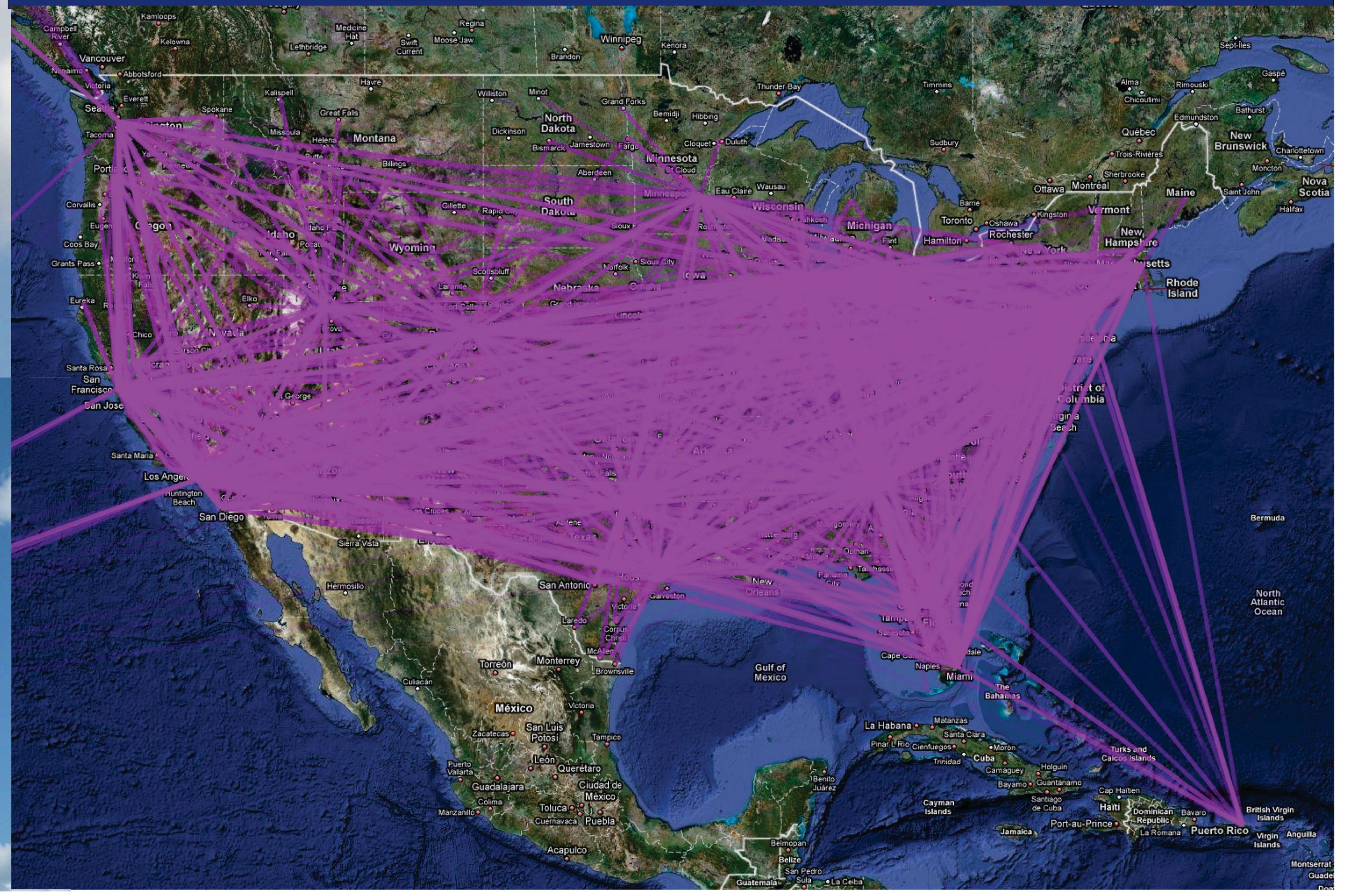
The data are provided by Research and Innovative Technology Administration (RITA) and Bureau of Transportation Statistics (BTS).
Arrival and departure details for 123 million commercial flights throughout the United States are recorded between October 1987 and December 2008, representing 29 commercial airlines and 3,376 airports. About 2.3 million flights were cancelled, 25 million flights were at least 15 minutes late.

Additional Sources

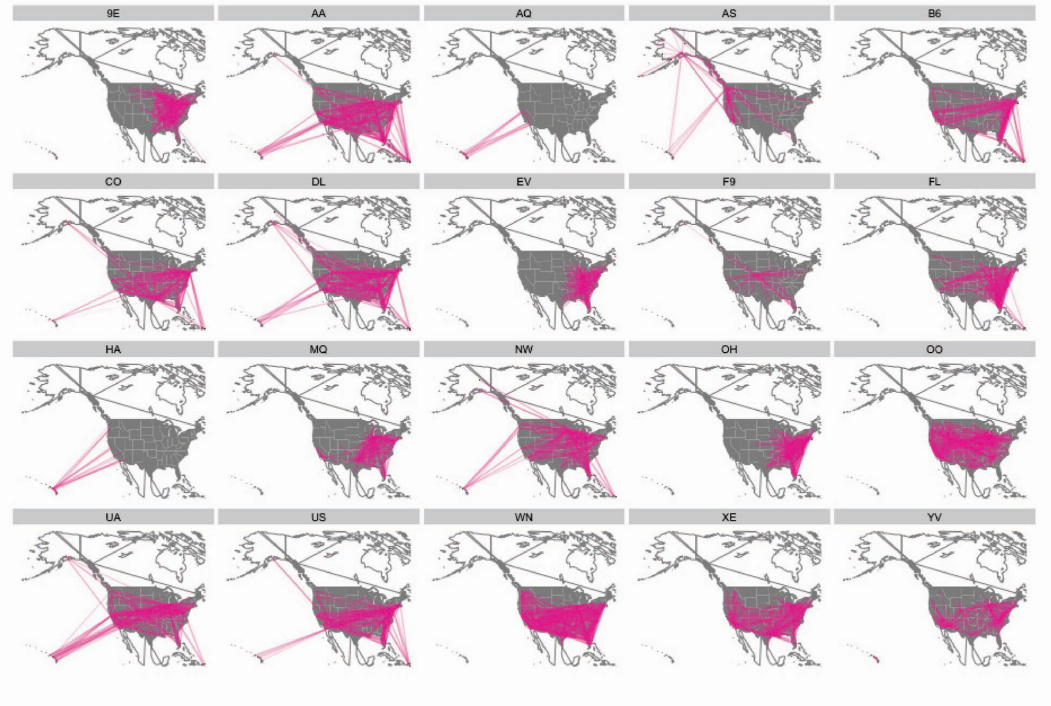
Additional BTS Data:

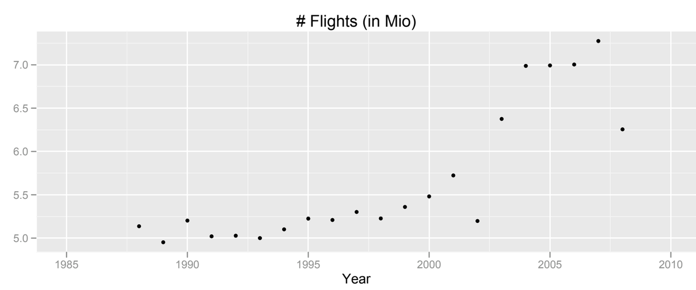
* Monthy fuel cost and consumption data by carrier
* Fleet information by carrier

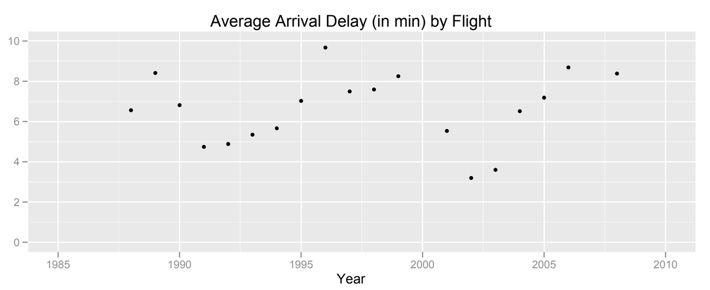
Hourly weather details for each airport from Weather Underground at http://www.wunderground.com



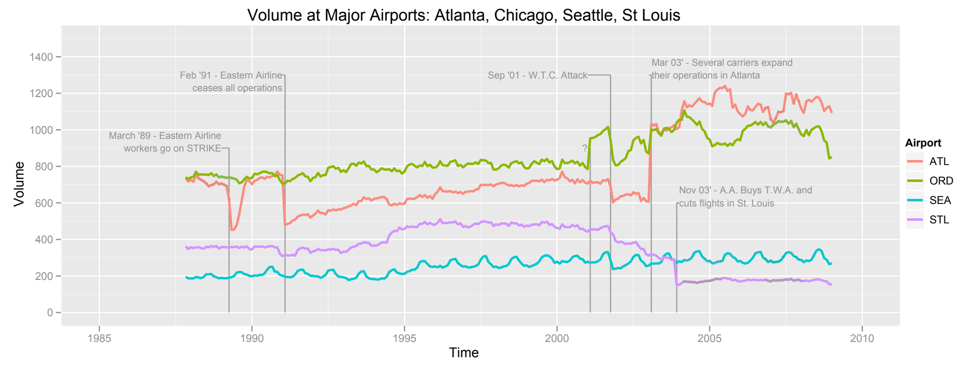
FLIGHTS OF `07
The maps above and below show all flights from 2007. Southwest Airlines (WN) operates without a hub system, the other airlines' hubs are prominent. Small carriers tend to operate locally.





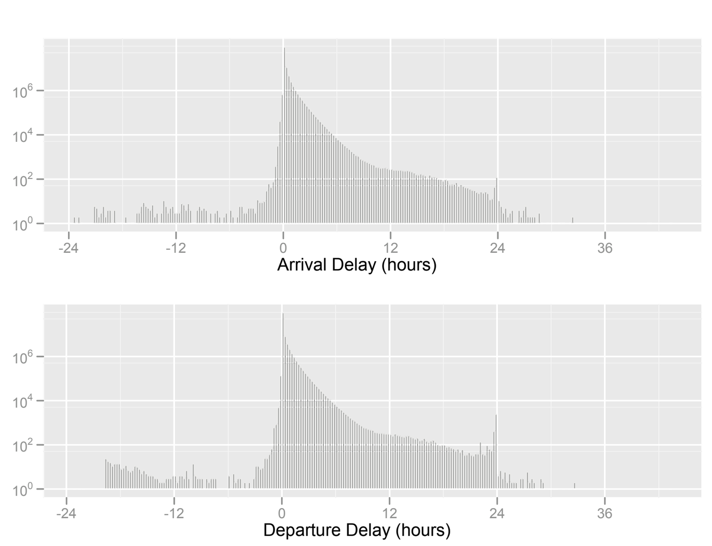


Flight volume is on the increase - dramatically so since 2000. Structural shifts (below) in the flight load for airports lead to minimal average delays (above, right) in `02 and `03. Delays have been on the increase since.

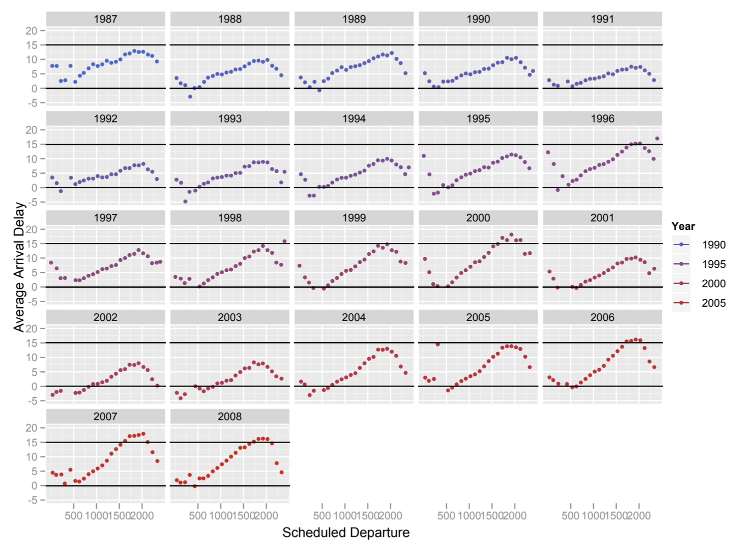


Large changes in airports' daily flight volume are triggered by different events, including strikes, FAA order, and seasonal fluctuations,

Delays

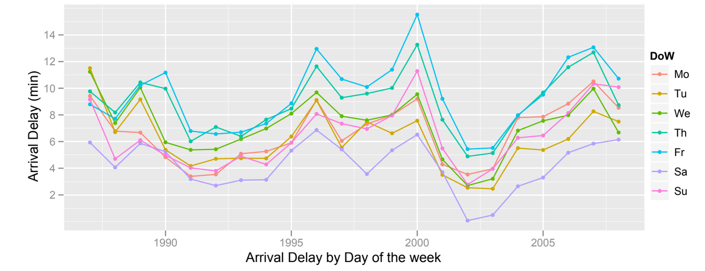


ARRIVAL & DEPARTURE
Most flights have 0 delay, with fewer and fewer flights having increasing delays. A secondary peak occurs at 24 hours suggests a limit of 24 hours delay is used by some carriers. Some data is likely incorrect, eg flights arriving 24 hours early.

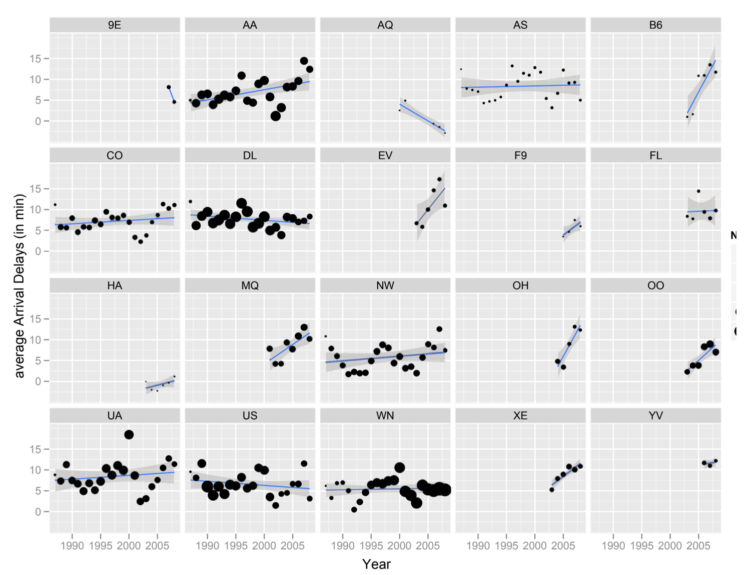


BY SCHEDULED DEPARTURE TIME & YEAR
Delays increase as day progresses. In 2001 they show an overall decrease, likely to be structural change, maybe FAA policy. Delays deteriorated again after 2003.

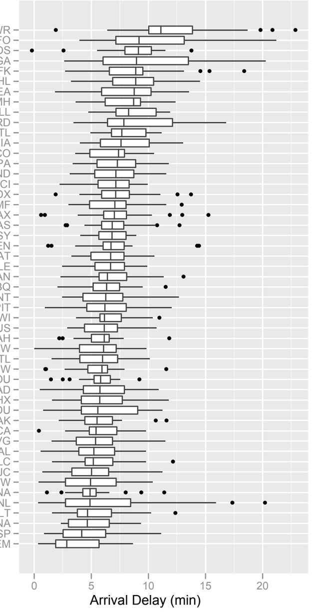
BY DAY OF WEEK
Best days to travel and avoid delays are Saturdays, and Tuesdays or Wednesdays. Fridays are bad for delays.

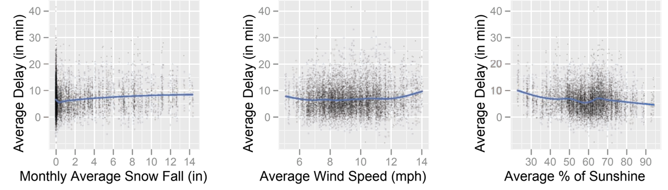


DELAYS BY CARRIER, YEAR
Increasing delays for small carriers, except for Aloha. Delta and US Airways are improving.



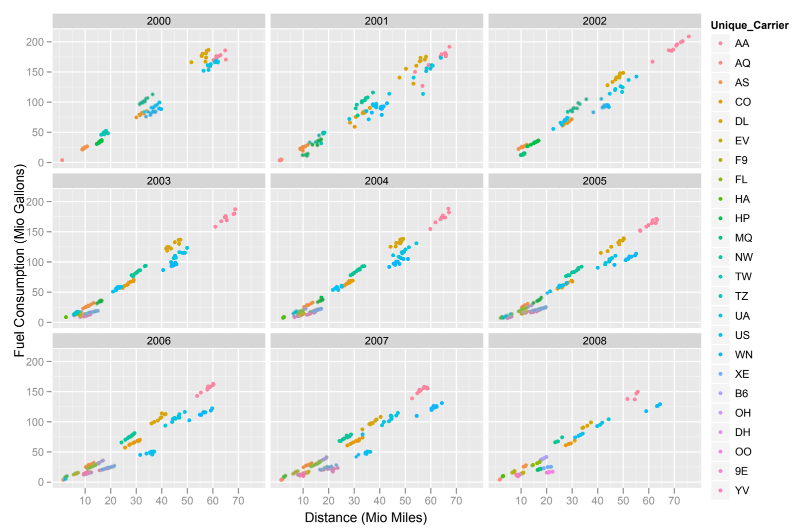
Carrier	Name
9E	Pinnacle
AA	American
AQ	Aloha
AS	Alaska
B6	JetBlue
CO	Continental
DL	Delta
EV	Atlantic
F9	Frontier
FL	AirTran
HA	Hawaiian
HP	America West
MQ	American Eagle
NW	Northwest
OH	Comair
OO	Skywest
UA	United
US	US Airways
WN	Southwest
XE	Expressjet
YV	Mesa



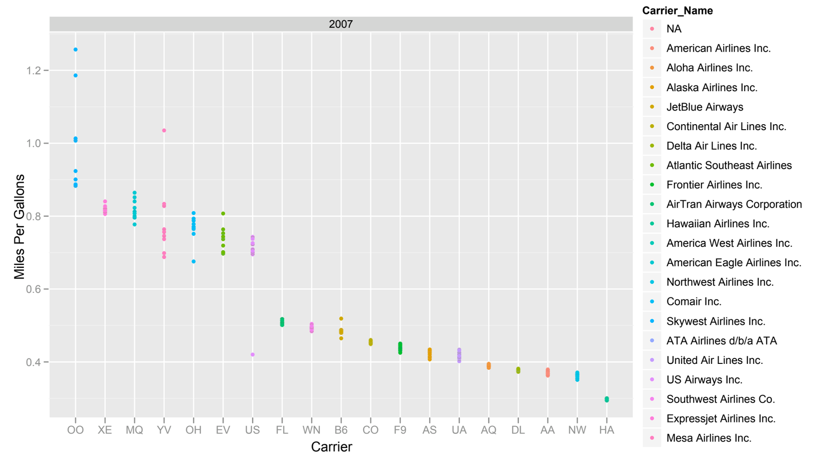


BY AIRPORT
EWR (Newark) is the worst. ORD (Chicago O'Hare) is not good, but also has high volume. DFW (Dallas-Fort Worth) is relatively good - high traffic but relatively small delay. Weather plays a huge role in delays - any kind of preciptation, high winds, or reduced visibility increases delays (scatterplots above)

Fuel Efficiency



FUEL USE
Consumption versus distance flown. Three groups of carriers are operating in 2000 - American Airlines is moving into the lead, Southwest is moving out of the middle group, closing up with AA and has overtaken AA on distance flown and efficiency 2007-8.



FUEL EFFICIENCY
Smaller carriers more efficient, probably due to use of smaller planes. Larger carriers American is one of the least efficient, Southwest is most efficient. Hawaiian Airlines very inefficient.

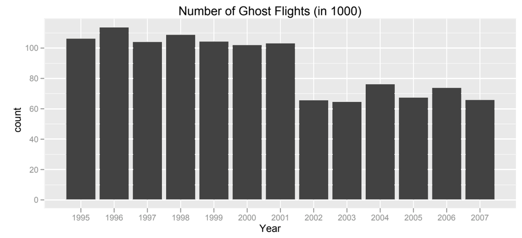
Ghosts of Flights

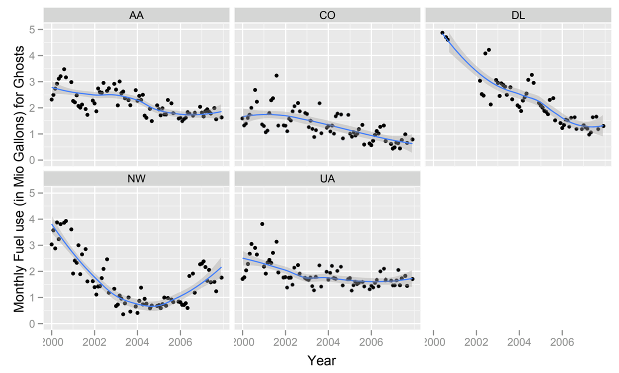
CAN WE SEE WHAT IS NOT THERE?
Planes have, for reasons such as maintenance, weather, or schedule fly empty between airports as so-called *Ghosts*. By tracking individual planes, we reveal their paths, including situations, where a plane lands in a different airport than where it takes off later, i.e. a ghost:

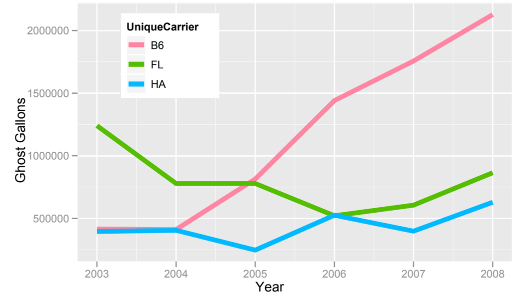
Example: US Airways Aircraft N-881 - Ghostflight from PIT to RIC (222 miles)

Year	Month	Day	DepTime	ArrTime	Origin	Dest	Diverted
1995	3	8	1102	1256	PIT	CVG	0
1995	3	8	1311	NA	CVG	PIT	1
1995	3	8	1913	2050	RIC	PIT	0
1995	3	8	2134	2300	PIT	MSY	0

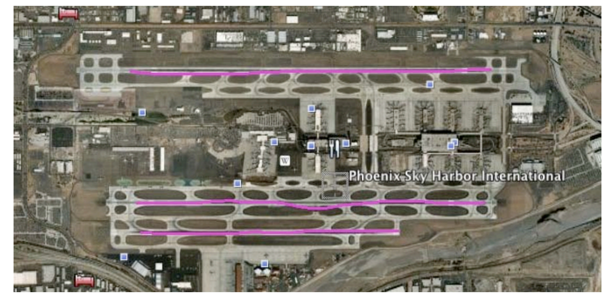
Ghost Flight Totals: over 1 million flights since 1995, with an average distance between airports of 1000 miles, corresponding to about 1.5 million gallons of fuel. Since 2001 the number of ghost flights is cut, with major airlines also decreasing the fuel consumption.
Smaller carriers are facing increasing costs from ghost flights.



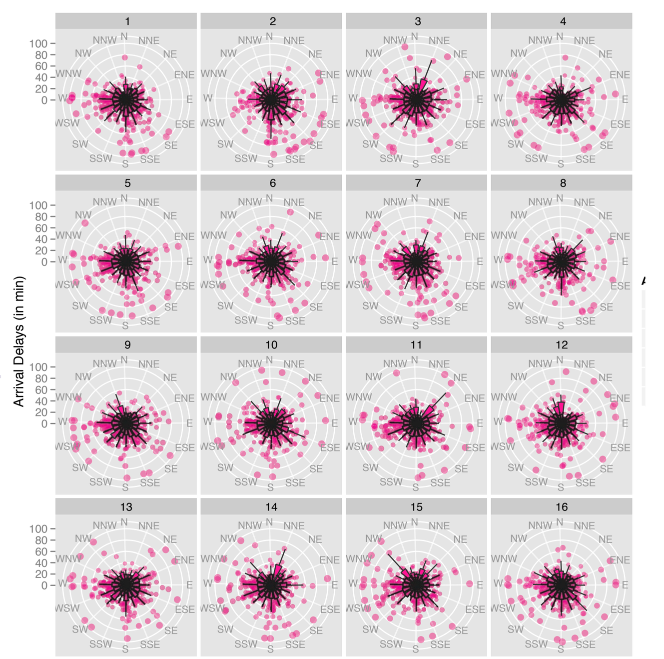




Crosswinds



All three runways in Phoenix are oriented East-West (see above). Do crosswinds affect air traffic?
On the right, arrival delays are plotted against permutations of wind direction (null plots). One plot shows the real data of the observed wind directions.



Can you spot which one?
If you can differentiate the true plot from the null plots, we can reject the null hypothesis of crosswinds being unrelated to arrival delays with a p-value of less than 0.0625 (= 1/16).

Believe it or not??

Racing Balloons?
Three of American Airlines' registered vehicles in the database are RAVEN hot air balloons. Based on the data, they cruise at an impressive average speed of 430 miles an hour. Fasten those hats!

I'm sorry, Sir, but your flight left 12h early'
According to the data, this was said to all passengers of 247 flights. Another 165 flights left at least 2h early.

"Within-City Hoppers"- if you're in a real time crunch
A total of 232,809 flights cover a distance of less than 50 miles. The shortest commercial flights occur between the New York airports La Guardia (LGA) and John F. Kennedy (JFK). The distance is 11 miles, which according to google.maps can be covered in 18 min by car, and according to data, takes 14 min of air time.

Tools

mySQL database on fast server (thanks to Ted Peterson)

R and packages
- ggplot2 (Hadley Wickham)
- DBI, RMySQL (David James, Jeff Horner)

Google Earth

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